PHOENIXTEKK

Workplace Marketplace · Intune Solutions

BitLocker Drive Encryption

Implementation Solution Guide

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Executive Summary — BitLocker Drive Encryption (Microsoft Intune)

Phoenixtekk Workplace Marketplace · Intune Solutions

A laptop left in a taxi is only a breach if its drive is readable. **BitLocker** encrypts the whole disk so a lost or stolen device exposes nothing — and managed through Microsoft Intune, it turns on **silently** on TPM-equipped devices, with every **recovery key escrowed to Microsoft Entra ID** so the help desk can always recover access.

This is a complete, **deploy-it-today** package for implementing BitLocker Drive Encryption with Microsoft Intune, written for an administrator with no prior BitLocker-management experience.

What you get

- A plain-language **Solution Overview** of how Intune-managed BitLocker works.
- A **Requirements Guide** with verify / implement-if-missing for every prerequisite.
- An **Implementation Guide** of precise, click-by-click Intune steps in the required order.
- Five **PowerShell scripts** for readiness discovery, encryption validation, tenant reporting, remediation, and recovery-key retrieval.
- **Testing** and **Troubleshooting** guides for confident rollout.

What this delivers

An Intune **disk encryption policy** that requires **XTS-AES 256** encryption of the OS (and fixed) drives, **silently** on TPM devices, with **recovery keys automatically saved to Entra ID** and full encryption reporting.

The Phoenixtekk deployment model

Assess → Configure → Pilot → Validate → Expand.

Outcome

Fleet-wide full-disk encryption with escrowed recovery keys — device-loss data protection and a compliance cornerstone — deployable by a single administrator and validated before rollout.

1. Solution Overview — BitLocker Drive Encryption

What BitLocker is

BitLocker is the full-volume encryption built into Windows. It encrypts the operating-system drive (and optionally fixed data drives) so the data is unreadable without the correct key. Managed through Microsoft Intune, encryption is enforced by policy, enabled **silently** on devices with a TPM, and every **recovery key is escrowed to Microsoft Entra ID**.

How it works

1. An Intune **disk encryption policy** defines the encryption requirements (method, drives, protectors).

2. On a targeted device, BitLocker enables — **silently** when a TPM is present and the user has no action to take.

3. The OS drive is protected by the **TPM** (optionally plus a startup PIN); the encryption uses **XTS-AES 256**.

4. The **recovery key is automatically backed up to Entra ID**, visible to admins in Intune (device → Recovery keys) and Entra (device → BitLocker keys).

5. Intune reports each device's encryption status for compliance and audits.

What you can require

• **OS drive** encryption with TPM protector (and optional startup PIN), XTS-AES 256.

• **Fixed data drive** encryption with auto-unlock.

• **Recovery options** — store recovery key in Entra ID, hide recovery options from the user, client-driven recovery password rotation.

• **Silent enablement** for standard users on TPM devices.

Benefits

• **Device-loss protection** — a lost/stolen device exposes no readable data.

• **Silent and zero-touch** — no user action on TPM hardware.

• **Always recoverable** — keys escrowed to Entra ID; the help desk can recover access.

• **Compliance cornerstone** — satisfies "require encryption" in device compliance.

Limitations (be honest with buyers)

• **Silent** enablement requires a **TPM** and a supported, properly configured device.

• Devices without a TPM can encrypt but require a startup PIN/USB key (not silent).

• The first full encryption takes time (used-space-only is faster); plan pilot timing.

• Self-encrypting drives / certain firmware quirks may need hardware-specific handling.

The standard Phoenixtekk deployment model

> ***Create the disk encryption policy → require XTS-AES 256 + Entra key escrow + silent enablement → assign to a pilot → validate encryption + key escrow → expand.*** *Phoenixtekk pattern:* ***Assess → Configure → Pilot → Validate → Expand.***

2. Requirements Guide — BitLocker Drive Encryption

Verify **every** requirement before configuring. Each entry: **Description · Why · Verify · Implement-if-missing.** Category-wide prerequisites are in `global-requirements-intune-solutions.md`.

2.1 Licensing

Description: Intune (Plan 1); BitLocker is a Windows feature (no separate license). **Why:** Intune delivers and reports the disk encryption policy. **Verify:** M365 admin center → **Billing** → **Licenses**. **Implement:** Assign M365 E3/E5 or Business Premium (Intune included).

2.2 Windows edition

Description: Windows 10/11 **Pro, Enterprise, or Education**. **Why:** BitLocker is not available on Windows Home. **Verify:** `winver` / **Settings** → **System** → **About** → **Edition**. **Implement:** Upgrade Home editions to Pro/Enterprise.

2.3 TPM (for silent enablement)

Description: TPM 1.2+ (2.0 recommended), enabled in UEFI. **Why:** Silent BitLocker enablement and the strongest protection require a TPM. **Verify:** `Get-Tpm` (`TpmPresent : True`, `TpmReady : True`); or run `BITLK-Discovery.ps1`. **Implement:** Enable TPM/PTT/fTPM in UEFI. Devices without TPM can still encrypt with a startup PIN/USB key (not silent).

2.4 UEFI / Secure Boot (recommended)

Description: UEFI firmware with Secure Boot. **Why:** Modern boot integrity improves BitLocker protection and pairs with health attestation. **Verify:** `Confirm-SecureBootUEFI`. **Implement:** Switch firmware to UEFI and enable Secure Boot (may require reinstall if legacy BIOS/MBR).

2.5 Microsoft Entra ID (key escrow)

Description: Devices Microsoft Entra joined (or hybrid joined). **Why:** Recovery keys are escrowed to the device's Entra object. **Verify:** `dsregcmd /status` → `AzureAdJoined : YES`. **Implement:** Join devices to Microsoft Entra.

2.6 Device enrollment

Description: Devices enrolled in Intune. **Why:** The policy and reporting require enrollment. **Verify:** Intune → **Devices** → **All devices**. **Implement:** Enable automatic enrollment; enroll devices.

Checklist

- Intune licensing assigned
- Windows Pro/Enterprise/Education
- TPM present + ready (for silent) — `BITLK-Discovery.ps1`
- UEFI + Secure Boot (recommended)

- Devices Microsoft Entra joined (key escrow)
- Devices enrolled in Intune
- Pilot Entra group created

Global Requirements — Intune Solutions (category-level)

These requirements apply to **every** Phoenixtekk solution in the **Intune Solutions** category. They are published once at the category level so individual listings stay small. Confirm these before deploying *any* Intune solution. Each entry: **Description · Why · Verify · Implement-if-missing.**

G1. Microsoft Entra ID tenant

Description: An active Microsoft Entra ID (Azure AD) tenant. **Why:** It is the identity backbone for every cloud and Intune-managed resource. **Verify:** Sign in to `https://entra.microsoft.com` as an administrator. **Implement:** Create a Microsoft 365/Azure subscription, which provisions a tenant.

G2. Microsoft Intune subscription

Description: An Intune-bearing license plan. **Why:** Intune delivers all configuration, compliance, and security policies. **Verify:** `https://intune.microsoft.com` loads; **Tenant administration** → **Tenant status** shows active. **Implement:** Assign Microsoft 365 E3/E5, Business Premium, or standalone Intune Plan 1.

G3. Administrative roles

Description: Appropriate Entra/Intune admin roles for the operator. **Why:** Creating policies and enrollment settings requires elevated rights. **Verify:** Entra → **Roles and administrators**; confirm **Intune Administrator**. **Implement:** Assign **Intune Administrator**; reserve Global Administrator for tenant-wide changes.

G4. Device enrollment in Intune

Description: Targeted Windows devices are enrolled and manageable. **Why:** Policies only reach enrolled devices. **Verify:** Intune → **Devices** → **All devices**; the device appears and is **compliant**. **Implement:** Enable **Automatic Enrollment** and join devices to Entra ID.

G5. Pilot security group

Description: A Microsoft Entra security group for pilot/targeted users or devices. **Why:** Every Phoenixtekk solution is piloted before fleet rollout. **Verify:** Entra → **Groups**; the pilot group exists. **Implement:** Entra → **Groups** → **New group** → **Security** → add members.

G6. Supported Windows baseline

Description: Windows 10 21H2+ or Windows 11 with current updates. **Why:** Modern Intune features assume a patched, supported OS. **Verify:** `winver`, or Intune → device → **Hardware**. **Implement:** Deploy

Update rings or upgrade to Windows 11.

G7. Network access to Microsoft cloud endpoints

Description: Outbound HTTPS to Microsoft 365, Entra, and Intune service URLs. **Why:** Enrollment, policy sync, and authentication depend on these endpoints. **Verify:** From a device, reach `https://login.microsoftonline.com` and `https://enrollment.manage.microsoft.com`. **Implement:** Allow Microsoft's published Office 365/Intune URL & IP ranges through proxy/firewall.

The Phoenixtekk deployment model (all solutions): Assess → Configure → Pilot → Validate → Expand.

3. Implementation Guide — BitLocker Drive Encryption

Follow these steps **in order**, in the **Microsoft Intune admin center** (`https://intune.microsoft.com`).

> ***Model:*** *A disk encryption policy requiring XTS-AES 256, Entra key escrow, and silent enablement, assigned to a pilot group.*

Step 1 — Create the pilot group

1. Entra → **Groups** → **New group** → Security → `BitLocker-Pilot` → add pilot devices/users. **Create**.

Step 2 — Create the disk encryption policy

1. Intune → **Endpoint security** → **Disk encryption** → **Create policy**.
2. **Platform:** Windows. **Profile: BitLocker**. **Create**.
3. **Name:** `BitLocker - Pilot`. **Next**.

Step 3 — Configure BitLocker settings

On **Configuration settings**:

1. **BitLocker – Base Settings**

• **Enable full disk encryption for OS and fixed data drives** = **Yes**.

• **Require storage cards to be encrypted (mobile only)** = not applicable.

• **Hide prompt about third-party encryption** = **Yes**.

• **Allow standard users to enable encryption during Autopilot** = **Yes**.

• **Configure client-driven recovery password rotation** = **Enabled for Microsoft Entra joined devices**.

1. **BitLocker – Fixed Drive Settings**

• **Fixed drive recovery** = **Configure** → **Save BitLocker recovery information to Microsoft Entra ID** = **Yes**; **Store recovery information before enabling** = **Yes**.

• **Block write access to fixed drives not protected by BitLocker** = optional.

1. **BitLocker – OS Drive Settings**

• **Encryption for OS drives** = **Require**.

• **BitLocker with non-compatible TPM chip** = block (or allow PIN if you support non-TPM).

• **OS drive recovery** = **Configure** → **Save BitLocker recovery information to Microsoft Entra ID** = **Yes**; **Store recovery information before enabling** = **Yes**; **Hide recovery options during BitLocker setup** = **Yes**.

• **Encryption method** = **XTS-AES 256-bit**.

1. **BitLocker – Removable Drive Settings** = as required. Click **Next**.

Step 4 — Assign to the pilot group

1. **Assignments** → **Add groups** → `BitLocker-Pilot`. **Next** → **Create**.

Step 5 — Apply on a pilot device

1. On a pilot device, sync policy (**Settings** → **Accounts** → **Access work or school** → **Info** → **Sync**).
2. On TPM devices encryption begins **silently**; otherwise the user is prompted for a startup PIN.

Step 6 — Validate the pilot

1. Run `BITLK-Validation.ps1` → confirms the OS drive is encrypting/encrypted with the right method and a recovery protector exists.
2. In Intune → device → **Recovery keys**, confirm the **recovery key is escrowed**.
3. Run `BITLK-TenantReport.ps1` → confirms encryption status across the pilot group. Complete the **Testing Guide**.

Step 7 — Expand to production

1. Add a production group to the policy **Assignments**; roll out in waves.
2. Confirm key escrow for each wave (a device that encrypts without an escrowed key is a recovery risk).
3. Re-validate with `BITLK-Validation.ps1` / `BITLK-TenantReport.ps1`.

Order of operations

```
Pilot group → Disk encryption policy → Base/Fixed/OS settings (XTS-AES 256 + Entra escrow + silent) → Assign → Apply → Validate encryption + escrow → Expand
```

5. Testing Guide — BitLocker Drive Encryption

Run after the pilot policy applies (Implementation Step 6) and after each production wave.

Test 1 — Policy applied

Do: Intune → **Endpoint security** → **Disk encryption** → `BitLocker - Pilot` → **Device status**. **Expected:** Pilot devices show **Succeeded** (no Error).

Test 2 — OS drive is encrypted

Do: On a pilot device run `BITLK-Validation.ps1` or `manage-bde -status`. **Expected:** OS drive **Protection On**, **Percentage Encrypted = 100%**, **Encryption Method = XTS-AES 256**.

Test 3 — Silent enablement (TPM devices)

Do: Observe a TPM device after policy sync — no user prompt. **Expected:** Encryption begins automatically with no user interaction.

Test 4 — Recovery key escrowed to Entra

Do: Intune → device → **Recovery keys** (or Entra → device → **BitLocker keys**). **Expected:** A recovery key is present for the device.

Test 5 — Recovery key retrieval works

Do: Run `BITLK-Get-RecoveryKey.ps1 -DeviceName` (Graph). **Expected:** The recovery password is returned for an authorized admin.

Test 6 — Compliance reflects encryption

Do: Intune → device → **Compliance** (with a "require encryption" compliance rule). **Expected:** The encrypted device is **Compliant** for the encryption requirement.

Test 7 — Tenant report

Do: Run `BITLK-TenantReport.ps1`. **Expected:** A breakdown of encrypted / encrypting / not-encrypted devices.

Pass/fail gate

Gate	Requirement
Pilot pass	Tests 1–5 pass for the pilot group (escrow confirmed for every device)
Wave pass	Tests 2, 4, 7 pass before the next wave

6. Troubleshooting Guide — BitLocker Drive Encryption

Use `BITLK-Validation.ps1` / `BITLK-Discovery.ps1` for device state and `BITLK-Remediation.ps1` for safe fixes.

A. Encryption doesn't start

Symptom	Cause	Remediation
Policy "Succeeded" but not encrypting	No TPM, or TPM not ready	Run `BITLK-Discovery.ps1`; enable TPM in UEFI; for non-TPM, configure a startup PIN
"Encryptable volume" error	Secure Boot/UEFI or pre-provisioning state	Ensure UEFI + Secure Boot; reboot; re-sync policy
Standard user can't enable	"Allow standard users to enable during Autopilot" not set	Enable that base setting

B. Recovery key not escrowed (critical)

Symptom	Cause	Remediation
Encrypted but no key in Entra	"Store recovery info before enabling" off, or device not Entra joined	Set escrow-before-enable; confirm `AzureAdJoined : YES`; run `BITLK-Remediation.ps1` to back up the key
Key missing for older devices	Encrypted before policy	`BITLK-Remediation.ps1` backs up the existing recovery password to Entra

C. Wrong encryption method

- Device shows AES-128 / different method → it was encrypted before the policy. Decrypt and re-encrypt with XTS-AES 256, or accept per risk; new devices follow the policy.

D. TPM problems

- `Get-Tpm` shows not ready → enable/clear TPM in UEFI (coordinate: clearing TPM affects existing keys — have the recovery key first).
- TPM "reduced functionality" → clear via `tpm.msc` (with recovery key in hand).

E. Diagnostics quick reference

```
manage-bde -status                       # encryption %, method, protection state
(Get-BitLockerVolume -MountPoint C:).KeyProtector   # protectors (TPM, RecoveryPassword)
Get-Tpm                                  # TPM readiness
manage-bde -protectors -get C:           # view recovery password ID
```

Back up an existing key to Entra: `BackupToAAD-BitLockerKeyProtector -MountPoint C: -KeyProtectorId .`

F. When to escalate

If a device has a TPM ready, is Entra joined, the policy shows Succeeded, and encryption still won't start or the key won't escrow after remediation, collect `manage-bde -status` output and the BitLocker event logs and open a Microsoft support case.

7. Best Practices — BitLocker Drive Encryption

- **Escrow the key before enabling.** Always set "Store recovery information before enabling" — a device that encrypts without an escrowed key is a recovery disaster waiting to happen.
- **Pilot first.** Confirm silent enablement and key escrow on a small group before fleet rollout.
- **Require XTS-AES 256.** The modern, strong method for fixed internal drives.
- **Prefer silent enablement on TPM devices.** No user friction; reserve startup PINs for high-security scopes or non-TPM hardware.
- **Enable client-driven key rotation** for Entra-joined devices so recovery passwords refresh after use.
- **Pair with compliance.** Use a "require encryption" compliance rule so unencrypted devices are flagged (and blocked via Conditional Access).
- **Mind the TPM before clearing it.** Clearing/resetting a TPM can invalidate keys — always have the recovery key first.
- **Encrypt fixed data drives too**, with auto-unlock, for full coverage.
- **Report continuously.** Track encryption coverage and escrow with `BITLK-TenantReport.ps1`.
- **Document recovery procedure.** Make sure the help desk knows how to retrieve a recovery key (`BITLK-Get-RecoveryKey.ps1` / Intune device → Recovery keys).

8. Deliverables Summary — BitLocker Drive Encryption

Documentation

Document	Purpose
Executive Summary	Business case and what's in the box
1. Solution Overview	What Intune-managed BitLocker is, how it works, benefits, limitations
2. Requirements Guide	Every prerequisite with Verify / Implement-if-missing
Global Requirements (Intune Solutions)	Category-wide prerequisites, published once
3. Implementation Guide	Exact, ordered, click-by-click Intune steps
5. Testing Guide	Validation procedures with pass/fail gates

Document	Purpose
6. Troubleshooting Guide	Field issues and remediation
7. Best Practices	Field-proven recommendations

PowerShell scripts

Script	What it does
`BITLK-Discovery.ps1`	Readiness: TPM, Windows edition, UEFI/Secure Boot, current encryption state
`BITLK-Validation.ps1`	Confirms the OS drive is encrypted (XTS-AES 256) with a recovery protector
`BITLK-TenantReport.ps1`	Tenant-wide encryption status + recovery-key presence (Microsoft Graph)
`BITLK-Remediation.ps1`	Safe remediation; enables encryption and backs up the recovery key to Entra
`BITLK-Get-RecoveryKey.ps1`	Retrieves a device's BitLocker recovery key for authorized admins (Microsoft Graph)

What the buyer receives

- This implementation guide as a branded **Word** document and **PDF**.
- All five **PowerShell scripts** in a single ZIP.
- A repeatable process following the Phoenixtekk model: **Assess** → **Configure** → **Pilot** → **Validate** → **Expand.**

Outcome

Fleet-wide full-disk encryption with escrowed recovery keys — device-loss protection and a compliance cornerstone.

Appendix A — PowerShell Scripts

Included in the accompanying ZIP. Each script's header documents full usage.

BITLK-Discovery.ps1

BITLK-Get-RecoveryKey.ps1

BITLK-Remediation.ps1

BITLK-TenantReport.ps1

BITLK-Validation.ps1